# Transfer Learning Conversational QA from Expanded Extractive QA

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## ABSTRACT

Extractive Question Answering (QA), the task of selecting a span to answer a question asked about a passage, is a long studied task. Recently, conversational question answering (ConvQA) emerged as a promising new research area. The main differences between the two tasks are that there are multiple turns and additional flexibility of the questions, such as coreferences, topic shift, or drill down among others. Even though there is a moderate amount of task overlap, there seems to be no studies on transfer learning ConvQA from extractive QA. Yet another problem of extractive QA is that answer spans are generally short, which are not desirable in a conversational setting. Thus, we propose a method to expand the ground-truth answer spans of extractive QA datasets (such as SQuADv2 and NewsQA) and use these altered datasets as an intermediate fine-tuning task before final fine-tuning on ConvQA. Despite being such a simple approach, this intermediate fine-tuning does improve downstream performance on QuAC as opposed to models that have not done so. We have released our code at https://github.com/gzhihongwei/convqa_from_qa for further study on this phenomenon.

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## 1 INTRODUCTION

In conversational systems, the ability to answer questions is important to give the user confidence that the conversational agent understands speech and is intelligent enough to correctly answer the question posed. In general, many questions asked by users to these conversational agents can be either non-factoid, which require reasoning and the ability to synthesize many sources, or factoid, which can be directly answered by reading a relevant document. In this work, we don't address non-factoid questions and limit factoid question answering to the task-oriented dialogue (TOD) setting.

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In the context of the Alexa Prize TaskBot Challenge 2021, which is the first prize to spur research in conversational systems that help with step-by-step tasks, the factoid question answering task becomes quite a bit more focused: towards reading comprehension.

Why does this form of question answering capture the needs for the TaskBot? For a given conversation, a user can choose either a recipe-based task or a DIY task. Once such a task is selected (given question identification and classification are working appropriately), most questions will either be directly about the task (in-task) or be a general question (open domain). If the question is correctly classified as in-task, then the answer to the user's question must be contained in the task's document text or in the bot's utterances. Thus, we can use the already said bot utterances along with the task document text as the context for a reading comprehension style question answering. So in order to answer such in-task questions, we will need to focus on reading comprehension style question answering.

Previous question answering datasets (and hence models) have usually focused on two specific forms: short-form and long-form. In our setting within TODS, both forms are problematic. Short-form answers, like the name suggests, are usually very short sequences of tokens that a model has determined to be the answer, which could either be a noun phrase or be part of a noun phrase. When the answer returned is the latter, the answer provides a poor user experience in the conversational setting because the bot is not providing the answer in context. On the other hand, long-form answers are usually on the sentence to paragraph level, which would also provide a poor user experience because the answers themselves are too long (to be understood and remembered in speech). Since no viable datasets have been introduced to address question answering in TODS, we instead will look at the more general setting of conversational question answering (ConvQA).

In order to leverage previous extractive QA datasets that focus on short-form answers, we propose expanding the ground-truth answer span to its longest surrounding noun phrase. The motivation for such an approach is that the ground-truth answers for these short-form answer extractive QA datasets generally give the shortest answer span possible to answer the question. Like was said previously, this is not favorable in the conversational setting. When considering a constituency parse of the context that an answer span is selected from, we will continually expand the span until we hit the longest answer span that is a noun phrase.

This begs the question, **RQ:** does performance on downstream ConvQA benefit from intermediate fine-tuning on expanded extractive QA datasets? From our experiments, there is evidence to suggest that performance on ConvQA does benefit from this intermediate fine-tuning. Our contributions are as follows:

- scripts to expand answer spans in SQuAD [9, 10] and NewsQA [12];
- and open sourced code that is simple and validates the importance of intermediate fine-tuning on expanded extractive QA for higher downstream performance on ConvQA.

## 2 PRIOR WORK

One of the most recognized and well-known datasets that initially spurred modeling research on extractive QA was SQuAD, the Stanford Question Answering Dataset [9, 10]. Each question in SQuAD was generated by crowdworkers, who were given a Wikipedia document to generate the questions about. Although it can be argued that the questions generated in this manner are somewhat contrived, this sort of procedure aligns pretty well with the sorts of questions users make about the task they chose because users tend to be forgetful when conversing in speech. Yet another span selection reading comprehension dataset is NewsQA, which instead asks questions about news articles from CNN [12].

Nowadays, most models that are built to solve the reading comprehension task utilize Transformers [13] or its derivatives. Two modern approaches, Lan et al. [5] and Zhang et al. [14], are indicative of where the current trends in the state of the art lie.

As far as we are aware, there has been no prior work done in transfer learning ConvQA from extractive QA. In particular, expanded extractive QA has not been done in the literature, so this work is novel.

## 3 METHOD

Since our approach inspects the change of downstream performance on ConvQA with intermediate fine-tuning on expanded extractive QA, we will first introduce which datasets we used, rationalize why they were used, and how they were preprocessed before model training (§3.1). Then, we will describe how we expanded the ground-truth answer spans (§3.2). After, we will talk about the small changes that had to be made to RoBERTa [6] to make it suitable to the ConvQA dataset we picked (§3.3). Finally, we will mention some implementation details (§3.4).

### 3.1 Datasets

In extractive QA, SQuAD [10] and NewsQA [12] are the most well known datasets that focus on identifying an answer span given a question and an answer context. More specifically, SQuADv2 [9] was selected instead of v1 because it added ~50,000 unanswerable questions. All in all, SQuADv2 and NewsQA had around ~125,000 question-answer pairs each with possibly unanswerable questions.

As for the downstream ConvQA dataset, QuAC [1] was chosen because it has a similar setting of extracting a span from a document and having some unanswerable questions. Additionally, there are two additional dialogue acts of continuation and affirmation. The former act is to teach models when to encourage the questioner to ask a follow up question. The latter act is to identify when an affirmation question is posed, thereby making the model learn what type of question is being asked.

*3.1.1 Dataset Preprocessing.* For both extractive QA datasets, we took the contexts and expanded answer spans with the character start indices if the question is answerable (used `None` otherwise for the answer). Afterwards, the HuggingFace tutorial for QA dataset preprocessing was used[1] to get the data ready for model training. For QuAC, we did a similar process except that up to the last $k$ question-answer turns were prepended to question $i$ to create the reformulated question. Formally,

$$q_i^* = [q_{i-k}; a_{i-k}; \ldots; q_i] \tag{1}$$

where $q_i$ is the question at turn $i$, $q_i^*$ is the reformulated question, and $a_{i-k}$ is the answer to the question at turn $i-k$.

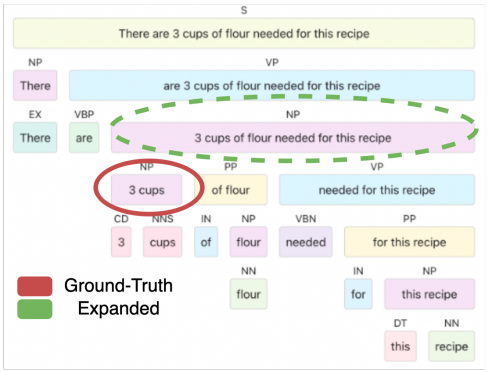


**Figure 1: Example of how the answer expanding pipeline works. The red oval is what the ground-truth answer span is labeled as and the green dotted oval is the expanded answer span that we are aiming to extract. Parsing and visualization of the parse tree is courtesy of Stern et al. [11], Kitaev and Klein [4], and Kitaev et al. [3].**

### 3.2 Expanding Extractive QA Ground-Truth Answers

In order to expand the ground-truth answer spans of the answerable questions in SQuAD and NewsQA, we need a constituency parse of the entire context. To do so, we leveraged the Berkeley Neural Parser [3, 4, 11] in combination with SpaCy [2]. We made sure to allow the parser to expand the text span if a constituent is not found initially. If after expanding there is still no constituent found, we simply copy over the ground truth answer span. Figure 1 illustrates the proposed method of expanding the answer span. Namely, we can traverse up the tree from the constituent given by the ground-truth answer span until we reach the largest noun phrase constituent in the parse tree.

Formally, Algorithm 1 is what is performed on both SQuADv2 and NewsQA to obtain SQuADv2$^\dagger$ and NewsQA$^\dagger$.

### 3.3 RoBERTa Baseline

In order to only isolate the performance difference of the intermediate fine-tuning on the data preprocessing rather than the model, we decided to use RoBERTa [6] base as the backbone. Since QuAC not only is concerned with identifying the answer span, we needed to add an additional head for predicting the dialogue states, such as identifying whether to add "yes," "no," or neither before the answer

[1] https://huggingface.co/course/chapter7/7?fw=pt

**Algorithm 1:** Expand ground-truth answer span

```
Data: Ground-truth answer span node n in neural
      constituency parse of surrounding context
Result: Largest answer span node largest in neural
        constituency parse such that n is in a subtree of
        largest and is a noun phrase
if n.label is None then
    largest ← n;
    return;
end
prev ← n;
while n.label is "NP" do
    prev ← n;
    n ← n.parent;
    if n is None then
        break;
    end
end
largest ← prev;
```

span and whether to encourage a follow up question (yes, maybe, or no). Since these two tasks are both 3 multi-class prediction problems, we added an additional linear layer on top of the `[CLS]` token to predict these two different dialogue state related subtasks.

If $\mathcal{L}_{\text{span}}$ is the loss for the span selection subtask, $\mathcal{L}_{\text{yesno}}$ is the loss for the affirmation subtask, and $\mathcal{L}_{\text{followup}}$ is the loss for the continuation subtask, then the total fine-tuning loss is given by

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{span}} + \mathcal{L}_{\text{yesno}} + \mathcal{L}_{\text{followup}}. \tag{2}$$

Specifically, every component of $\mathcal{L}_{\text{total}}$ is cross entropy loss. $\mathcal{L}_{\text{span}}$ is cross entropy on predicting positions within the maximum sequence length of RoBERTa, and the others are cross entropy on 3 classes. For simplicity, we simply added the separate losses together rather than a weighted combination.

At inference time, each turn is processed once at a time, the answer span is selected and the dialogue acts of confirmation and affirmation are predicted, then the reformulated questions for following turns are created based off of previous predicted answer spans.

### 3.4 Implementation Details

Regardless of the model, the maximum sequence length was capped at 386 tokens and document stride was set to 128 tokens. An effective batch size of 384 was used with a weight decay of 0.01. A linear warmup up to a learning rate of 3e-5 and linear decay with warmup ratio 0.2 over 5 epochs was used. Mixed precision training was enabled to speed up training time too. AdamW [7] was used to optimize every model.

For models that had intermediate fine-tuning, model weights were initialized from `deepset/roberta-base-squad2` on the HuggingFace Hub and trained with the above hyperparameters. Finally, the baseline model was initialized from `roberta-base` on the HuggingFace Hub.

**Table 1: Evaluation on the validation set of QuAC. † means that the dataset is preprocessed to have expanded answer spans. For all metrics, higher is better.**

| Models | F1 | HEQ-Q | HEQ-D | Yes/No | Follow up |
|---|---|---|---|---|---|
| RoBERTa | 39.6 | 37.2 | 1.4 | 86.6 | 55.5 |
| RoBERTa (SQuADv2) | 40.5 | 38.7 | 1.4 | 87.4 | 55.4 |
| + SQuADv2† | **42.6** | **40.5** | 2.1 | 87.4 | **55.8** |
| + NewsQA† | 41.5 | 39.4 | 1.9 | **87.4** | 54.9 |
| + SQuADv2† + NewsQA† | 41.3 | 39.3 | **2.4** | 86.9 | 54.5 |

When fine-tuning on QuAC, we chose to include up to the past $k = 3$ turns to reformulate a question at a given turn.

## 4 EXPERIMENTS

In order to answer **RQ**, we had to train a multitude of different models. For the baseline model, we initialized from the RoBERTa base checkpoint and fine-tuned on QuAC directly. Otherwise, we initialized from Deepset AI's checkpoint, fine-tuned on QuAC directly or performed intermediate fine-tuning on some combination of SQuADv2† and NewsQA† before the final fine-tuning on QuAC.

As can be seen in Table 1, initializing the model from an extractive QA checkpoint did better almost across the board. This suggests that single turn extractive QA is a helpful intermediate task for ConvQA, which is encouraging. When comparing this model that is trained with regular SQuADv2 knowledge with additional intermediate fine-tuning on our expanded answer-span datasets SQuADv2† and NewsQA†, it can be readily seen that performance for the latter is always superior in F1, HEQ-Q, and HEQ-D. It seems that the last two metrics of classification accuracy on the different dialogue acts were not very affected by the additional intermediate fine-tuning. This makes intuitive sense since the intermediate tasks do not have these dialogue acts.

## 5 SHORTCOMINGS

Even though QuAC [1] has been out since 2018, very few have publicized their fine-tuning code or the modeling is pre-Transformers[2]. I have compared my QuAC preprocessing code with Qu et al. [8][3], and they seem very similar. Since the model is more complicated and is written in TensorFlow without HuggingFace, it became very confusing to understand after the preprocessing step. Like I said before, I adapted the HuggingFace QA dataset preprocessing tutorial for QuAC preprocessing. This shouldn't be the problem, but I'm not quite sure. If the performance of our baseline was more inline with that of the leaderboard, the overall F1 would be between 10-20 points higher. It is unclear if this same behavior of intermediate fine-tuning on expanded extractive QA would have the same magnitude of improvement or not, so we can only speculate. So, we didn't have enough time to figure out the problem in the preprocessing/fine-tuning so that models could be submitted to the test server to evaluate on the test set.

[2] http://quac.ai
[3] https://github.com/prdwb/attentive_history_selection

One additional shortcoming is the lack of distinction between questions that are unanswerable given the entire context or questions that are answerable given the correct chunk of context, but the current context chunk does not contain the answer. Presumably, knowing the distinction that the question is answerable but not given the current context chunk could lead to improvement later on.

## 6 CONCLUSION

Despite the qualifications to the results given in §5, there is reason to believe that within closed domain ConvQA, intermediate fine-tuning on expanded extractive QA does improve downstream performance on metrics that could help provide a better user experience in the TODS setting, such as with HEQ-Q and HEQ-D, which are with comparison to humans. Perhaps something similar can be performed for more open-domain ConvQA tasks, but we will leave that to future work.

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